



Simulation and Control of Spin Waves in CrSBr Using FDTD and Laser-Cut Interfaces



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Abstract

Interest has recently risen in experimentally exploring the electrical, magnetic, and optical properties of CrSBr, a 2-dimensional (2D) semiconductor. Results have been found on the interaction between the spin waves this material hosts and interfaces due to changes in its thickness. Preliminary results have shown that wave-interface interaction varies by the type of excitation (i.e., line or point) and the interface geometry (i.e., straight line tilted at various angles). This leads to numerous setups that could be studied experimentally, but these experiments require significant time and monetary resources. Thus, developing a simulation based on well-established physical laws could allow for a preliminary testing of these different setups to inform which one might lead to the most intriguing experimental results. We approach this by using a Finite Difference Time Domain (FDTD) method. Additionally, as seen with other wave phenomena, different behavior arises from the interaction of different geometries (i.e., polygons, semicircles). This motivated the work on developing code to automate the laser cutting of the geometry of interest. Ultimately, this work assists in developing more options for experiments by using laser cutting procedures to obtain different geometries, as well as finding interesting phenomena by guiding experiments with simulations.

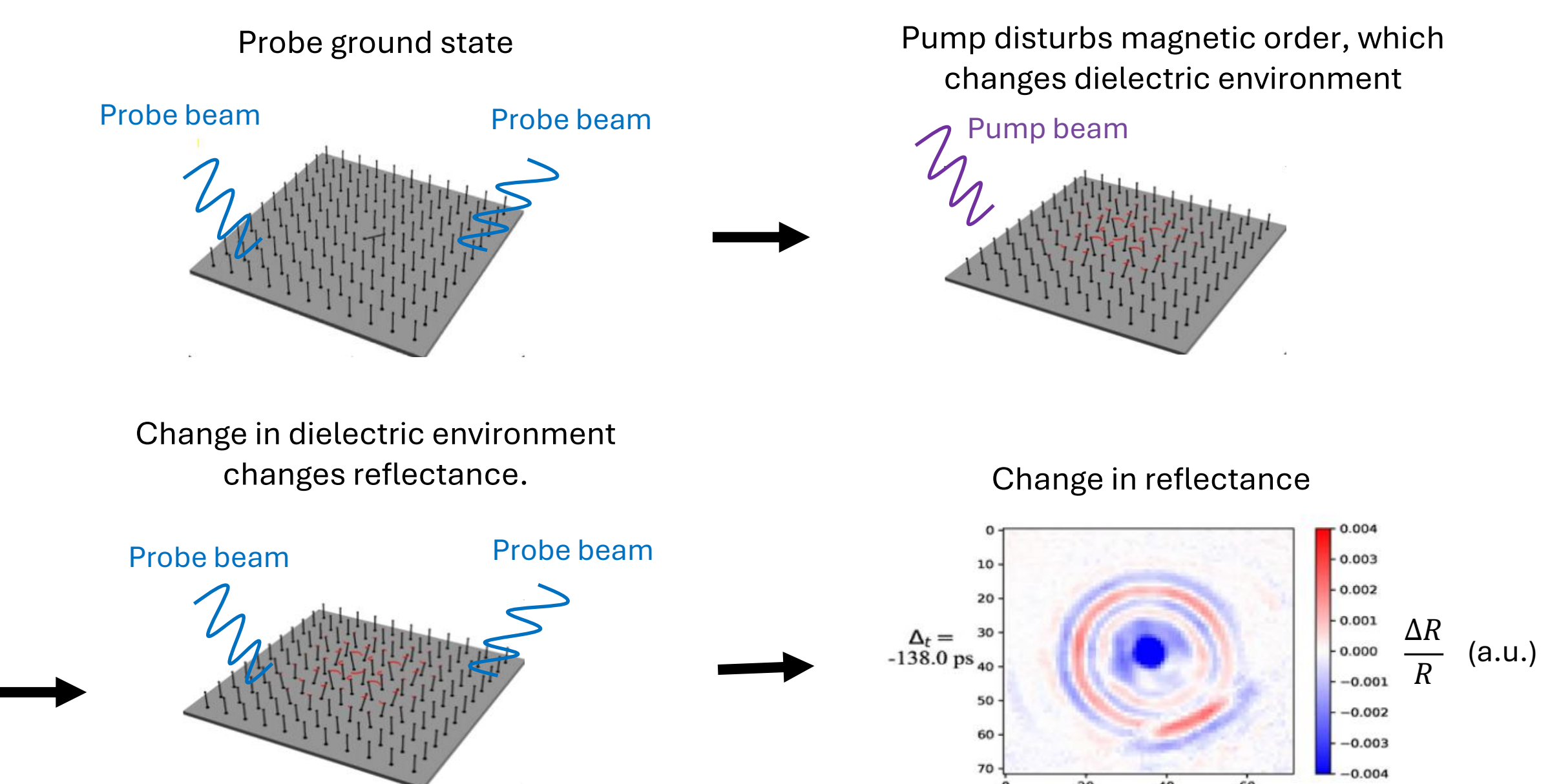
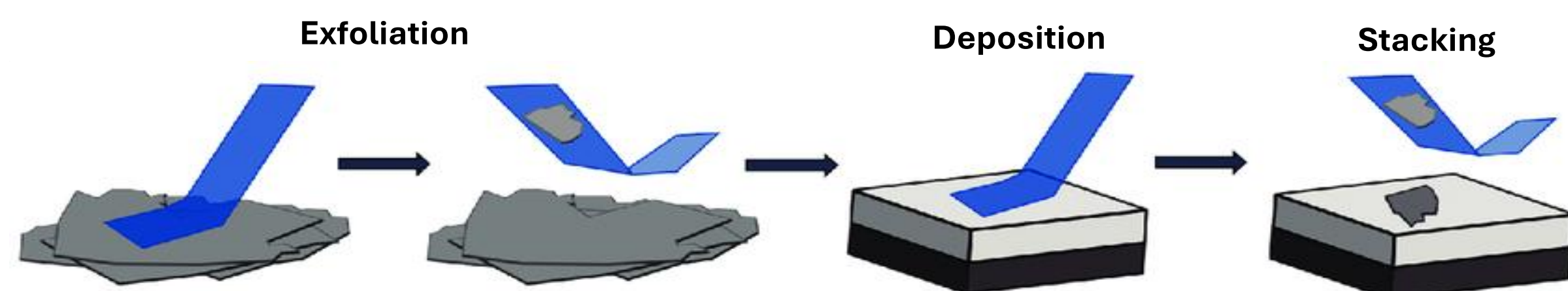
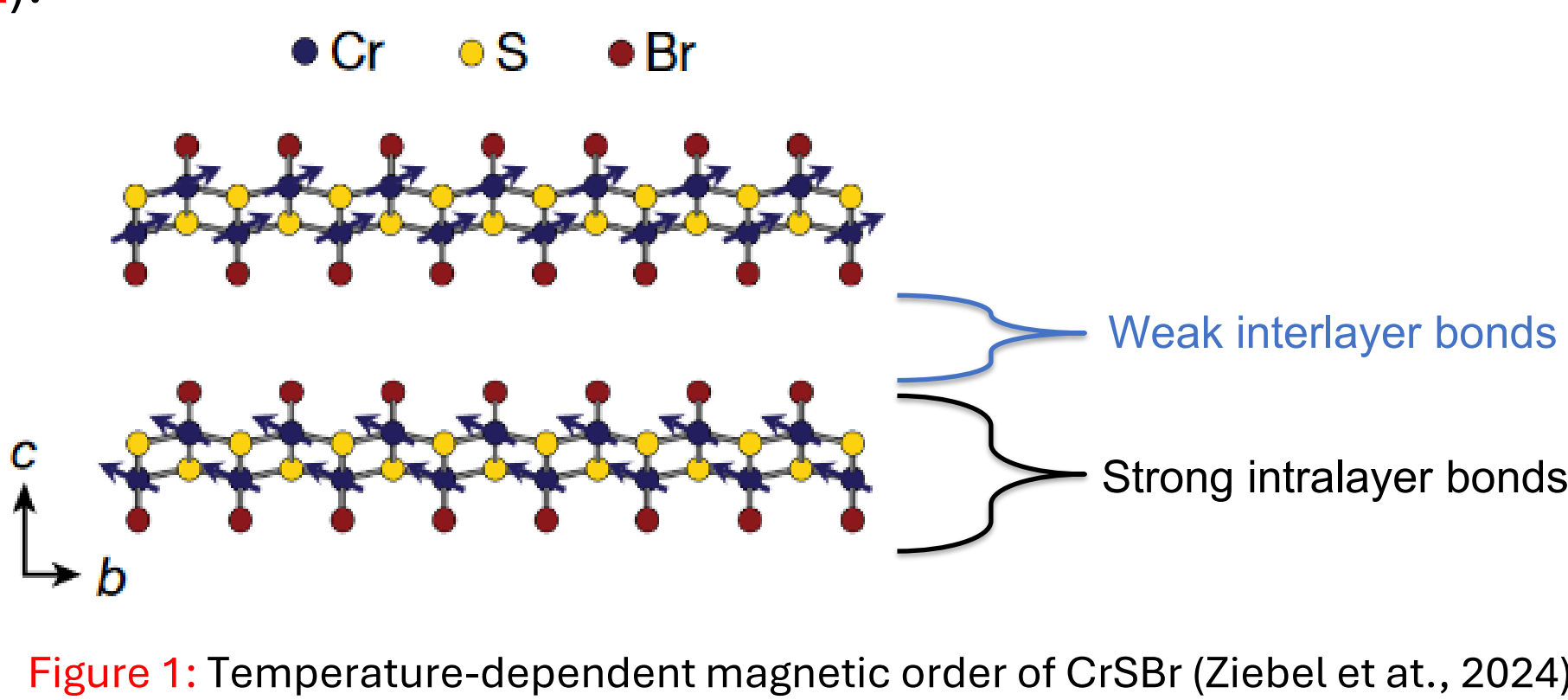
Sample Preparation and Data Collection

What are 2D materials?

These are a special family of materials that have strong intralayer bonding and comparatively weak interlayer interactions (Fig. 1), allowing the layer to be mechanically exfoliated by scotch tape (Fig. 2).

Spin wave generation and detection in CrSBr

The coupling between the optical and magnetic properties of CrSBr enables the detection and generation of spin waves through light (Fig. 3).



Lase-cutting System

Due to the nature of CrSBr crystal growth (rectangles), finding curved geometry that occurs naturally is extremely rare. This is why we're using a laser to fabricate the geometry of interest (Fig. 5a). The shape was drawn in the KLayout software (Fig. 5b). The file saved contains the corresponding indexing points of the polygon drawn shape. With a developed Python code, the motorized stage (Fig. 5c) of the laser-cutting system (Fig. 5d) was commanded to move according to the KLayout shape. With enough points and the right velocity settings, one can get near-perfect curves (Fig. 5e).

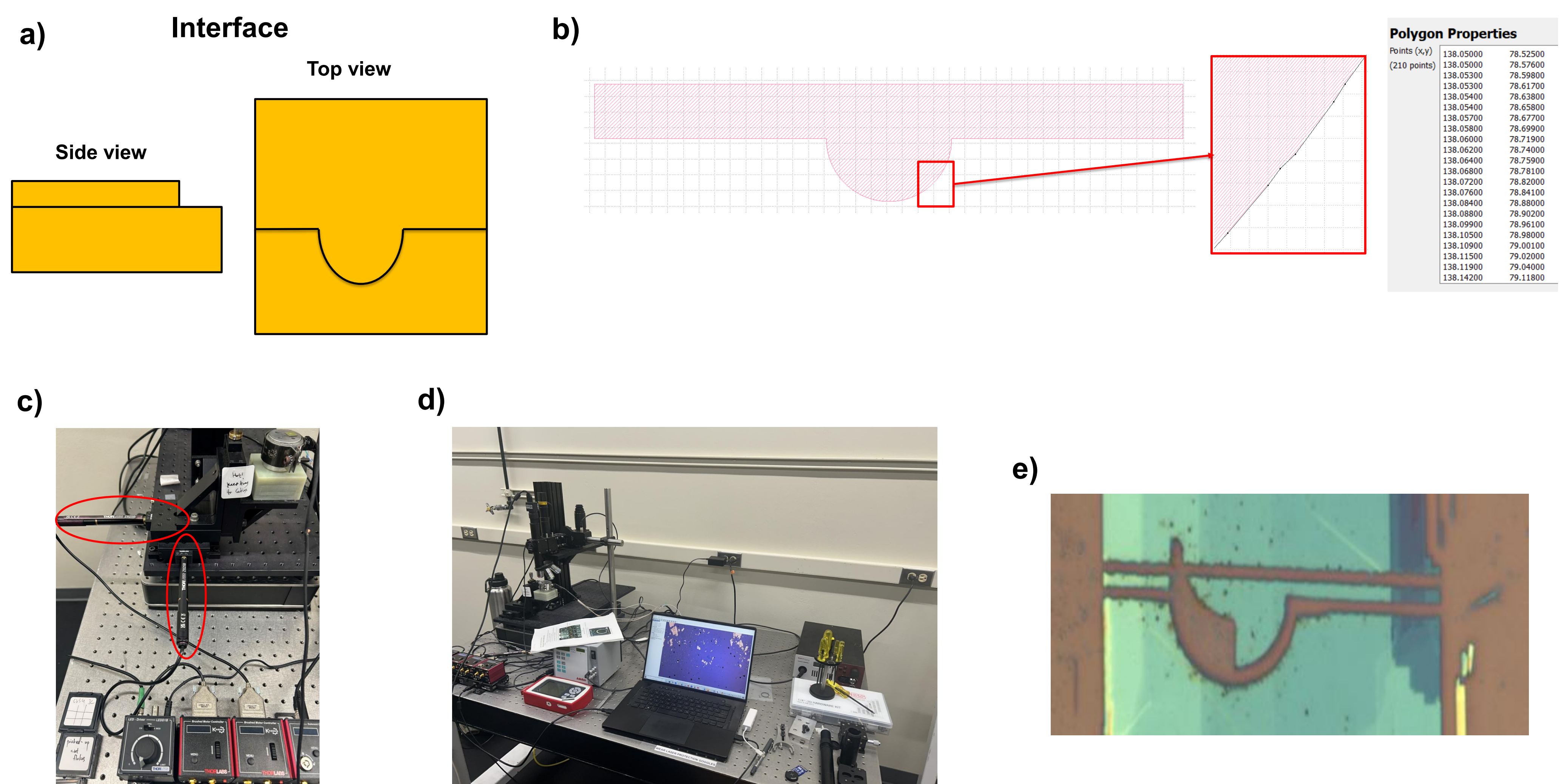


Figure 5: (a) Desired interface geometry. (b) KLayout drawing with its corresponding indices. (c) Motorized stage of the laser-cutting system. (d) Laser-cutting system. (e) Results of interface fabrication procedures.

Problem with Traditional Electronics

Technology applications

Traditional electronics rely on moving information encoded in charge, which produced heat proportional to current squared as it flows through a conductor (Eq. 1). So much heat is generated that this has driven data centers to be underwater (Fig. 4). Spin waves, by contrast, carry information without moving charge, reducing energy loss and enabling low-power computation.

New physics

The 2D material we are working with, CrSBr, hosts spin waves. These reveal rich nonlinear dynamics and offer a platform to potentially discover new physics.

About this project

We've observed that spin waves interact with interfaces due to changes in the thickness of CrSBr. To control and study this effect, we've developed a laser-cutting system to fabricate custom-shaped interfaces. Because of the vast number of possible geometries, we use simulations to guide and prioritize experiments.

Joule's Heating

$$P = I^2 R \quad (1)$$

Where P is power dissipated as heat, I is current, and R is the resistance of the material.



Figure 4: Microsoft data center being put underwater

Simulation Framework

From the experimental procedures given in Figure 3, point excitation experiments were conducted and compared to simulations (Fig. 6). The simulation was developed from the dispersion relation, Eq. (2), found by Bae et al., 2022. Substituting in the operators, a differential equation describing the propagation of spin waves in CrSBr is found (Eq. 3). This differential equation is solved numerically via Finite Difference Time Domain (FDTD) method. The simulation is essentially the numerical solution evolved over some time.

$$\omega(\mathbf{k}) = \alpha \sqrt{\beta + \frac{\gamma}{\delta + |\mathbf{k}|^2}} \quad (2)$$

Let $\omega = i \partial_t$ and $|\mathbf{k}|^2 = -\nabla^2$ and substitute to get

$$i \partial_t \psi = \alpha \sqrt{\beta + \frac{\gamma}{\delta - \nabla^2}} \psi$$

This corresponds to the following differential equation

$$(\delta - \nabla^2) \partial_t^2 \psi + \alpha^2 \beta (\delta - \nabla^2) \psi + \alpha^2 \gamma \psi = 0 \quad (3)$$

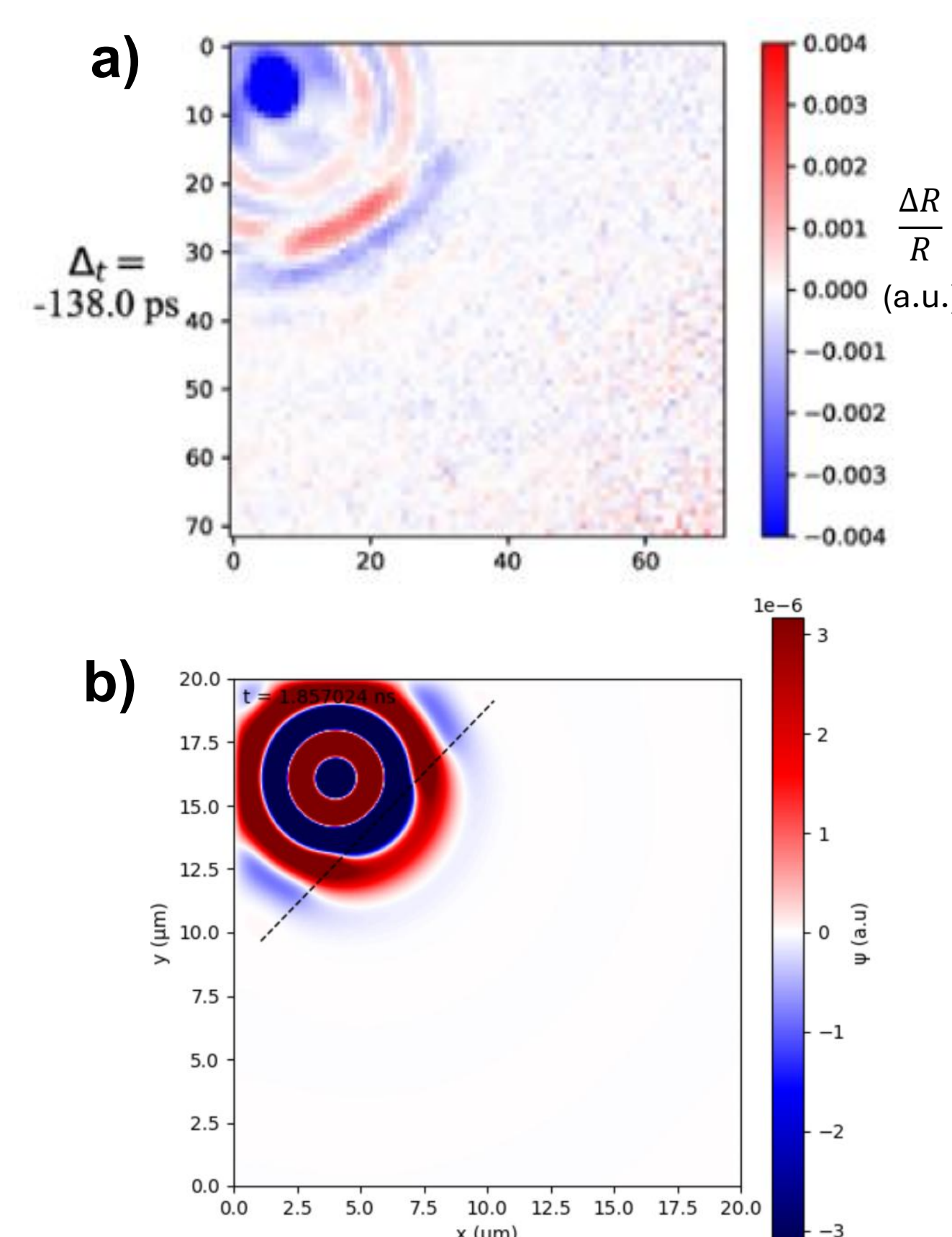


Figure 6: (a) Spin wave from experimental results. (b) Simulated spin wave. The dashed line in (b) represents the interface due to a change in the thickness of the material.